

Q1. How does a solar panel work? How is it tested and serviced?

Saranya R.

Palakad, Kerala

A1. The main component of a solar panel is a solar cell, also known as photovoltaic (PV) cell. It is a solidstate device, wherein a p-n junction with a very large light-sensitive area converts light energy into electricity by photovoltaic effect.

PV cells can be modeled as a current source in parallel with a diode and parasitic series and shunt resistance.

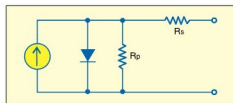


Fig. 1: Equivalent circuit of the solar cell

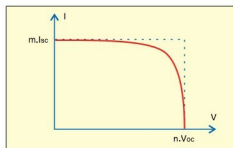


Fig. 2: I-V curve for solar modules

An equivalent circuit of the solar cell is shown in Fig. 1. When no light falls on the PV cell, it behaves like a diode. As the intensity of incident light increases, the PV cell generates more current.

To increase the utility of PV cells, dozens of individual cells are interconnected together in a sealed, weatherproof package called a module. To achieve the desired voltage and current, modules are wired in series and parallel into a PV array or solar panel.

Most solar cell parameters can be checked by simple I-V measurements. I-V characteristics of a typical solar module are shown in Fig. 2. Here 'n' is the number of cells connected in series, 'm' the number of cells connected in parallel, and I_{sc} and V_{oc} short-circuit current and open-circuit voltage values for individual cells.

V_{oc} is the maximum voltage that the

panel can produce when not connected to an electrical circuit or system, that is, when the current through the solar cell is zero. It is the maximum voltage available from the solar cell. V_{oc} can be measured with a meter directly in contact with the panel's terminals or the ends of its built-in cables.

I_{sc} is the current through the solar cell when the voltage across the solar cell is zero.

I-V testing systems include the light source, measurement electronics, computer and software needed to measure solar cell I-V curves. More details can be found in July 2011 issue of EFY.

Q2. How does a 12V battery charge?

Baliram Dagade

Pune

A2. A battery is a source of power that provides DC voltage.

It converts chemical energy into electrical energy. The battery stores energy when it is charged. It discharges when current is drawn from the load connected across its terminal. When the stored energy depletes beyond a limit indicated by a fall in terminal voltage, the battery has to be recharged using a charger.

A simple battery charger uses the mains 220V AC supply; it is stepped down to 15V, as about 14V is required to charge a 12V battery. The stepped-down AC voltage is rectified by a full-wave bridge rectifier, and filtered and regulated by a regulator IC. Voltage indicators and safety circuits prevent overcharging.



A 6V/12V battery charger

For battery charger circuits, you may refer to April 2008 and Jan. 2012 issues of EFY.

Q3. What is the working principle behind controlling the speed of a ceiling fan by remote control?

Padma Priya J.

Dinamalar Nagar

Ramanathapuram

A3. A ceiling fan is a single-phase AC machine and a remote control is a transmitter of infrared pulses. The working principle of fan speed control by remote therefore is based on controlling one of the parameters on which the speed of the fan depends. Physical characteristics of the motor cannot be changed. However, average power supplied to the load (in this case the ceiling fan) can be controlled. This can be done by using a triac.

Triacs are bidirectional AC switch-



Ceiling fan with remote

es, capable of controlling AC loads. These control AC power either through full-cycle switching or phase control of current to the load. Typical applications include motor-speed controls, heater controls and incandescent light controls.

The speed is controlled by controlling the firing angle of the triac. If the firing is delayed, power supplied will be less. This, in turn, will reduce the speed of the fan.

A simple circuit for speed control of a ceiling fan by remote was published in July 2006 issue of EFY.